

A Corrected Lower-Bound Note for Distribution-Free Contextual Dynamic Pricing

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Abstract

This note repairs, as far as possible, a proposed proof of a $\Omega(T^{2/3})$ minimax lower bound for the distribution-free contextual pricing model of Luo, Sun and Liu. The original proof idea embeds a non-contextual pricing problem by taking $\theta_0 = 0$. That embedding is legitimate, but the particular “tent bump” hard instances have a cusp at the optimal price and violate the quadratic revenue regularity assumption used in Luo–Sun–Liu’s paper. Consequently that proof does not solve the paper’s lower-bound problem.

The corrected result below is partial progress. Under the paper’s fixed Lipschitz and quadratic-revenue assumptions, a smooth parabolic-bump construction gives a rigorous

$$\inf_{\pi} \sup_I \mathbb{E}_I \mathcal{R}_T(\pi) \geq cT^{3/5}$$

lower bound, already in the non-contextual subclass. Moreover, for the non-contextual subclass this $T^{3/5}$ exponent is matched up to logarithms by a simple discretize-and-UCB algorithm, so a non-contextual embedding cannot by itself prove a $T^{2/3}$ lower bound under the fixed quadratic-revenue assumption. If the quadratic-revenue assumption is removed, the original tent construction can be made rigorous and gives the familiar $\Omega(T^{2/3})$ lower bound for the larger Lipschitz-only class. Thus the repaired argument is a solution for the larger Lipschitz-only formulation, but only partial progress for the Luo–Sun–Liu paper under its quadratic-revenue assumption.

1 Status and main conclusions

The model is the binary-feedback contextual pricing model

$$v_t = x_t^\top \theta_0 + z_t, \quad y_t = \mathbf{1}\{v_t \geq p_t\},$$

where θ_0 and the noise distribution are unknown. With survival function $D = 1 - F$, the expected revenue at context x and price p is

$$r_{\theta, D}(p, x) = pD(p - x^\top \theta).$$

Luo, Sun and Liu assume at least Lipschitz continuity of F , and their continuous-price regret bound uses the additional condition that, for

$$f_q(p) := p(1 - F(p - q)),$$

there is a fixed constant C such that

$$f_q(p_q^*) - f_q(p) \leq C(p_q^* - p)^2 \quad \text{for all feasible prices } p \tag{1}$$

whenever $q = x^\top \theta_0$. I call this the *quadratic-revenue* condition.

The attempted $T^{2/3}$ lower bound uses revenue functions with tent-shaped peaks. If a tent peak is centered at s , then for small $\delta > 0$ its revenue loss satisfies

$$R(s) - R(s + \delta) = a\delta.$$

This is incompatible with (1), because $a\delta \leq C\delta^2$ for all sufficiently small δ would force $C \geq a/\delta \rightarrow \infty$. Hence the tent proof cannot be a proof under a fixed quadratic-revenue constant.

The corrected finite-scale statements are the following.

Theorem 1 (What is solved and what is not). *For fixed problem-class constants, the following hold.*

- (a) *In the larger Lipschitz-only class, the non-contextual embedding gives a rigorous $\Omega(T^{2/3})$ lower bound. Together with known Lipschitz-only algorithms, this identifies the Lipschitz-only time exponent up to logarithms.*
- (b) *In the Luo–Sun–Liu class with the additional quadratic-revenue condition (1), the same strategy gives a rigorous $\Omega(T^{3/5})$ lower bound, not $\Omega(T^{2/3})$.*
- (c) *In the non-contextual subclass satisfying (1), a grid-UCB policy has regret $\tilde{O}(T^{3/5})$. Therefore no lower bound that uses only the non-contextual subclass can prove $\Omega(T^{2/3})$ for the fixed-constant quadratic-revenue class.*

Thus this note gives partial progress, not a solution of the original Luo–Sun–Liu lower-bound problem under the quadratic-revenue condition.

The rest of the note proves these statements. The constants below are not optimized; the point is that they are independent of T .

2 Model, subclasses and notation

Prices are restricted to $[0, 3]$. The restriction is harmless for lower bounds, since a lower bound on a smaller feasible price set remains a lower bound for any larger bounded price set after rescaling. In the non-contextual subclass we put $\theta_0 = 0$ and take an arbitrary fixed context sequence, so that $v_t = z_t$ and

$$R_D(p) := pD(p)$$

is the one-period expected revenue from price p . The pseudo-regret of a policy π on instance D is

$$\mathcal{R}_T(\pi, D) = \sum_{t=1}^T \left(\max_{p \in [0, 3]} R_D(p) - R_D(p_t) \right).$$

The interaction law induced by D is denoted by $\mathbb{P}_D$ or $\mathbb{P}_j$ when the instance is indexed by j .

For constants $L, C < \infty$, define $\mathcal{D}_{\text{Lip}}(L)$ to be the set of survival functions D on $\mathbb{R}$ such that D is nonincreasing, $D(-\infty) = 1$, $D(+\infty) = 0$, D is L -Lipschitz, and the corresponding valuations are bounded in $[0, 3]$. Define $\mathcal{D}_{\text{Quad}}(L, C)$ to be the subclass of $\mathcal{D}_{\text{Lip}}(L)$ for which every instance has an optimal price p^* and

$$R_D(p^*) - R_D(p) \leq C(p^* - p)^2, \quad p \in [0, 3]. \quad (2)$$

Embedding either class into the contextual model is done by taking $\theta_0 = 0$. If the contextual parameter set contains 0, every lower bound below is a valid contextual lower bound.

We use the standard inequalities

$$\text{TV}(P, Q) \leq \sqrt{\frac{1}{2} \text{KL}(P \| Q)} \quad (3)$$

and, for Bernoulli means $u, v \in [\eta, 1 - \eta]$,

$$\text{kl}(\text{Ber}(u), \text{Ber}(v)) \leq c_\eta (u - v)^2. \quad (4)$$

3 Why the tent construction cannot satisfy the quadratic condition

Let

$$\psi(u) = (1 - |u|)_+, \quad R_j(p) = b + ah\psi\left(\frac{p - s_j}{h}\right).$$

This is the revenue shape used by the attempted proof. The maximizer is $p^* = s_j$. For $0 < \delta < h$,

$$R_j(s_j) - R_j(s_j + \delta) = ah - a(h - \delta) = a\delta.$$

If (2) held with a fixed finite constant C , then $a\delta \leq C\delta^2$ for every sufficiently small δ , which is impossible. This is the exact mathematical obstruction: the construction has a locally linear peak, whereas the Luo–Sun–Liu assumption requires at most quadratic local loss around the optimizer.

A useful way to remember the obstruction is the following elementary consequence of (2).

Lemma 1 (Height-width restriction). *Suppose an instance in $\mathcal{D}_{\text{Quad}}(L, C)$ has a unique optimizer p^* . If $R_D(p^*) - R_D(p) \geq \Delta$ at some price p with $|p - p^*| \leq h$, then*

$$\Delta \leq Ch^2.$$

Thus a hard instance whose peak has width h can have height at most order h^2 when the quadratic constant is fixed.

Proof. This is immediate from (2). □

Lemma 1 explains the change of exponent. The Lipschitz case permits peaks of height order h , while the quadratic-revenue case permits only height order h^2 at width h .

4 A corrected lower bound under the quadratic-revenue assumption

This section proves a clean $T^{3/5}$ lower bound under fixed Lipschitz and quadratic-revenue constants. The construction is deliberately simpler than the nested bump towers used in some continuum-bandit lower bounds: for a horizon T , one smooth bump at one of $K \asymp h^{-1}$ possible locations is enough.

4.1 Construction of the instances

Fix $b = 1/4$ and choose $a \in (0, 1/64]$. Let

$$\phi(u) = (1 - u^2)_+.$$

The function ϕ has a quadratic peak:

$$1 - \phi(u) \leq u^2, \quad u \in \mathbb{R}. \quad (5)$$

For a bandwidth $h \in (0, 1/32]$, choose centers $s_1, \dots, s_K \in [5/4, 7/4]$ such that the intervals

$$S_j = [s_j - h, s_j + h]$$

are disjoint and $K \geq c_0/h$ for a universal constant $c_0 > 0$.

On $[1, 2]$, define

$$R_0(p) = b, \quad R_j(p) = b + ah^2\phi\left(\frac{p - s_j}{h}\right), \quad 1 \leq j \leq K, \quad (6)$$

and put $D_j(p) = R_j(p)/p$ on $[1, 2]$. Extend D_j to all real prices by

$$D_j(p) = \begin{cases} 1, & p \leq b, \\ b/p, & b \leq p \leq 1, \\ R_j(p)/p, & 1 \leq p \leq 2, \\ b(3-p)/2, & 2 \leq p \leq 3, \\ 0, & p \geq 3. \end{cases} \quad (7)$$

The formula is consistent at the joining points because all bumps are supported inside $[5/4 - h, 7/4 + h] \subset (1, 2)$.

Lemma 2 (Validity of the smooth instances). *For all sufficiently small fixed a and all $h \in (0, 1/32]$, the functions D_j defined by (6)–(7) are valid survival functions in $\mathcal{D}_{\text{Quad}}(L, C)$ for constants L, C independent of h, T and j . Moreover, for $j \geq 1$, the unique optimal price is $p_j^* = s_j$ and the optimal revenue is $b + ah^2$.*

Proof. The boundary behavior is immediate from (7): $D_j(p) = 1$ for $p \leq b$, $D_j(p) = 0$ for $p \geq 3$, and D_j is continuous. On $[b, 1]$ and $[2, 3]$ it is nonincreasing. It remains only to check monotonicity on $[1, 2]$. Where differentiable,

$$D'_j(p) = \frac{R'_j(p)p - R_j(p)}{p^2}.$$

Inside the support of the bump, $|R'_j(p)| \leq 2ah$, and outside the support $R'_j(p) = 0$. Since $p \leq 2$ and $R_j(p) \geq b$, on the increasing side of the bump we have

$$R'_j(p)p - R_j(p) \leq 4ah - b < 0$$

for the chosen constants; on the decreasing side it is even smaller. Hence D_j is nonincreasing. The displayed formulas also give a Lipschitz constant bounded independently of h .

The revenue on $[1, 2]$ is exactly R_j . On $[b, 1]$ it is equal to b , on $[0, b]$ it is equal to p , and on $[2, 3]$ it is $bp(3-p)/2 \leq b$. Thus the only revenue exceeding b is the bump in (6), whose unique maximum is at s_j with value $b + ah^2$.

Finally we check (2). If $p \in [1, 2]$, then by (5),

$$R_j(s_j) - R_j(p) = ah^2 \left[1 - \phi\left(\frac{p - s_j}{h}\right) \right] \leq a(p - s_j)^2.$$

If $p \notin [1, 2]$, then $R_j(p) \leq b$ and $|p - s_j| \geq 1/4$, so

$$R_j(s_j) - R_j(p) \leq b + a \leq 16(b + a)(p - s_j)^2.$$

Thus (2) holds with, for example, $C = a + 16(b + a)$. $\square$

4.2 Information bound

Let $\mathbb{P}_j$ be the joint law of the adaptive interaction under D_j , and let $\mathbb{P}_0$ denote the flat baseline instance. Define the central interval

$$C_j = [s_j - h/2, s_j + h/2]$$

and the number of times the policy prices in that interval,

$$N_j = \sum_{t=1}^T \mathbf{1}\{p_t \in C_j\}.$$

Lemma 3 (The correct interval is rarely found on average). *There are universal constants $\gamma_0, c_1 > 0$ such that if $h = \gamma T^{-1/5}$ with $0 < \gamma \leq \gamma_0$ and T is large enough, then*

$$\frac{1}{K} \sum_{j=1}^K \mathbb{E}_j N_j \leq \frac{T}{2}.$$

Proof. Since the intervals C_j are disjoint,

$$\frac{1}{K} \sum_{j=1}^K \mathbb{E}_0 N_j \leq \frac{T}{K} \leq chT. \quad (8)$$

For every j ,

$$\mathbb{E}_j N_j \leq \mathbb{E}_0 N_j + T \text{TV}(\mathbb{P}_j, \mathbb{P}_0).$$

By Pinsker's inequality and Jensen's inequality,

$$\frac{1}{K} \sum_{j=1}^K \text{TV}(\mathbb{P}_j, \mathbb{P}_0) \leq \sqrt{\frac{1}{2K} \sum_{j=1}^K \text{KL}(\mathbb{P}_0 \| \mathbb{P}_j)}. \quad (9)$$

The adaptive KL chain rule gives

$$\text{KL}(\mathbb{P}_0 \| \mathbb{P}_j) = \mathbb{E}_0 \sum_{t=1}^T \text{kl}(\text{Ber}(D_0(p_t)), \text{Ber}(D_j(p_t))).$$

The two Bernoulli parameters differ only when $p_t \in S_j$. On S_j they are bounded away from 0 and 1, and

$$|D_j(p) - D_0(p)| \leq ah^2.$$

Thus (4) implies

$$\text{KL}(\mathbb{P}_0 \| \mathbb{P}_j) \leq c_1 a^2 h^4 \mathbb{E}_0 \sum_{t=1}^T \mathbf{1}\{p_t \in S_j\}.$$

Because the support intervals S_j are disjoint,

$$\frac{1}{K} \sum_{j=1}^K \text{KL}(\mathbb{P}_0 \| \mathbb{P}_j) \leq c_2 a^2 h^4 \frac{T}{K} \leq c_3 a^2 T h^5. \quad (10)$$

Combining (8)–(10),

$$\frac{1}{K} \sum_{j=1}^K \mathbb{E}_j N_j \leq chT + T\sqrt{c_3 a^2 T h^5 / 2}.$$

With $h = \gamma T^{-1/5}$ and γ sufficiently small, the right-hand side is at most $T/2$ for all sufficiently large T . $\square$

Theorem 2 (Quadratic-revenue lower bound). *There exists a fixed class $\mathcal{D}_{\text{Quad}}(L, C)$ and a constant $c > 0$ such that, for all sufficiently large T ,*

$$\inf_{\pi} \sup_{D \in \mathcal{D}_{\text{Quad}}(L, C)} \mathbb{E}_D \mathcal{R}_T(\pi, D) \geq cT^{3/5}.$$

Consequently the Luo–Sun–Liu contextual problem also has a $\Omega(T^{3/5})$ lower bound under the fixed quadratic-revenue condition.

Proof. For instance j , if $p \notin C_j$, then $|p - s_j| \geq h/2$ if $p \in [1, 2]$, while outside $[1, 2]$ the revenue is at most b . In either case,

$$R_j(s_j) - R_j(p) \geq \frac{a}{4}h^2.$$

Therefore

$$\mathbb{E}_j \mathcal{R}_T \geq \frac{a}{4}h^2(T - \mathbb{E}_j N_j).$$

Averaging over j and using Lemma 3,

$$\max_{1 \leq j \leq K} \mathbb{E}_j \mathcal{R}_T \geq \frac{1}{K} \sum_{j=1}^K \mathbb{E}_j \mathcal{R}_T \geq \frac{a}{8}Th^2.$$

With $h = \gamma T^{-1/5}$ this is $cT^{3/5}$. The contextual lower bound follows by taking $\theta_0 = 0$ and any fixed context sequence. $\square$

5 A matching non-contextual upper bound under the quadratic condition

The previous theorem is not just an artifact of a weak proof. In the non-contextual subclass, the fixed quadratic-revenue condition really changes the exponent from $2/3$ to $3/5$.

Proposition 1 (Grid-UCB upper bound in the non-contextual subclass). *For every fixed L, C , the non-contextual pricing problem over $\mathcal{D}_{\text{Quad}}(L, C)$ has a policy with regret*

$$\sup_{D \in \mathcal{D}_{\text{Quad}}(L, C)} \mathbb{E}_D \mathcal{R}_T \leq C'T^{3/5}(\log T)^{2/5}$$

for a constant C' depending only on the fixed class constants.

Proof. Discretize $[0, 3]$ by a uniform grid G with mesh η . Then $|G| \leq 4/\eta$. Let p^* be an optimal price and let $g \in G$ satisfy $|g - p^*| \leq \eta$. By (2),

$$R_D(p^*) - R_D(g) \leq C\eta^2.$$

Run any gap-independent UCB algorithm on the finite arm set G with bounded rewards $p_t y_t \in [0, 3]$. Its regret relative to the best grid arm is $O(3\sqrt{|G|T \log T})$. Therefore

$$\mathbb{E}_D \mathcal{R}_T \leq CT\eta^2 + C_1\sqrt{T \log T/\eta}.$$

Choosing $\eta = (\log T/T)^{1/5}$ gives the asserted bound. $\square$

Corollary 1 (Non-contextual embedding barrier). *A lower-bound proof that only uses the non-contextual subclass $\theta_0 = 0$ cannot establish a $\Omega(T^{2/3})$ minimax lower bound for the fixed-constant class $\mathcal{D}_{\text{Quad}}(L, C)$.*

6 What the original tent construction proves without quadratic regularity

For comparison, I include the corrected Lipschitz-only version of the original argument. This is the part of the attempted proof that was essentially right. It proves a result for the larger class in which (2) is not imposed.

Theorem 3 (Lipschitz-only lower bound). *There is a fixed Lipschitz class $\mathcal{D}_{\text{Lip}}(L)$ and a constant $c > 0$ such that, for all sufficiently large T ,*

$$\inf_{\pi} \sup_{D \in \mathcal{D}_{\text{Lip}}(L)} \mathbb{E}_D \mathcal{R}_T(\pi, D) \geq cT^{2/3}.$$

Proof. The proof is identical to that of Theorem 2, except that the parabolic bump $ah^2\phi((p-s_j)/h)$ is replaced by the tent bump $ah\psi((p-s_j)/h)$, where $\psi(u) = (1-|u|)_+$. The survival functions are again $D_j(p) = R_j(p)/p$ on $[1, 2]$, with the same monotone extension outside $[1, 2]$. For sufficiently small fixed a , the monotonicity argument remains valid because $|R'_j| \leq a$ and $2a < b$.

Now the central interval miss cost is order h , while the Bernoulli mean perturbation is also order h . Hence the average KL is bounded by cTh^3 rather than cTh^5 . Taking $h = \gamma T^{-1/3}$ makes the average KL small, while the regret lower bound is order $Th = T^{2/3}$. As shown in Section 3, however, the resulting revenue peak is locally linear and violates (2) for every fixed finite quadratic constant. $\square$

7 Interpretation

The proposed $T^{2/3}$ proof was close to a correct lower bound for a different problem, but not for the Luo–Sun–Liu paper as written. The exact point of failure is the peak geometry: a tent peak has height proportional to its width, whereas the quadratic-revenue assumption forces the height of a width- h peak to be at most order h^2 . This changes the information-theoretic balance from

$$Th^3 \asymp 1, \quad Th \asymp T^{2/3}$$

in the Lipschitz-only class to

$$Th^5 \asymp 1, \quad Th^2 \asymp T^{3/5}$$

in the fixed quadratic-revenue class.

Accordingly, the repaired statement is not a solution of the original Luo–Sun–Liu lower-bound problem if that problem asks for a fixed-constant $\Omega(T^{2/3})$ lower bound under the quadratic-revenue condition. The note establishes rigorous partial progress: a valid $\Omega(T^{3/5})$ lower bound under the paper’s smoothness condition, a proof that the non-contextual route is blocked at $\tilde{O}(T^{3/5})$, and a rigorous $\Omega(T^{2/3})$ lower bound for the broader Lipschitz-only formulation.

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